

Internet Control Message Protocol (ICMP)

The Internet Control Message Protocol (ICMP) is a network layer protocol used by routers and devices to **send error messages and operational info**.

- It helps troubleshoot connection problems, though it does not send user data or use a connection handshake like TCP.
- ICMP is commonly used by network tools such as **ping** or **traceroute**.
- **Layer 3 Protocol**: Operates at the network layer, directly inside IP packets.
- **Connectionless**: Does not establish a session or perform a handshake before sending messages.
- **No Ports**: Does not use port numbers to target specific software applications on a device.

Key Functions of ICMP

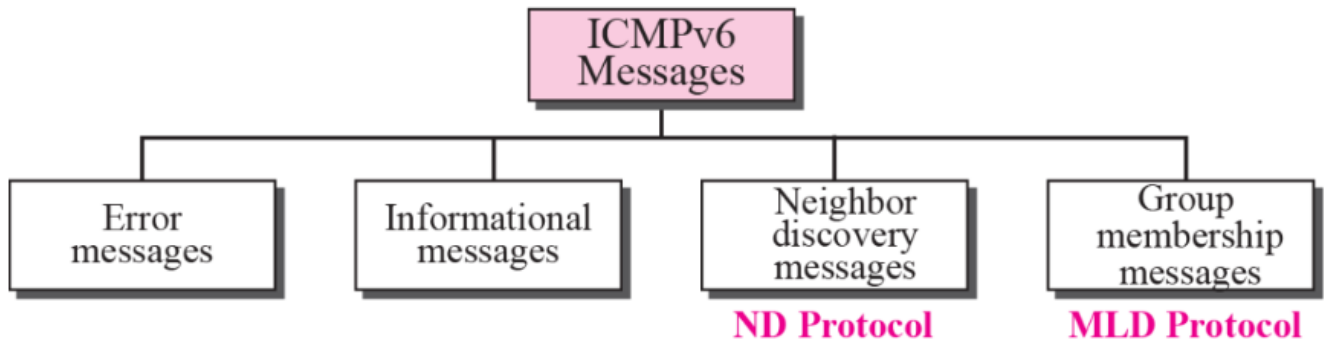
- **Error Reporting**: Tells the sender when a packet fails to reach its destination, such as when a host is down or a router drops a packet.
- **Network Diagnostics**: Powers tools like ping and traceroute to test if a device is online and measure latency.
- **Time Exceeded**: Sends a message when a packet's Time-to-Live (TTL) counter hits zero.
- **Destination Unreachable**: Reports when a network or port cannot be reached.

Note : ICMP does not carry normal user data ICMP is not normally used to carry:

- Web pages
- Videos
- Emails
- Files
- Chat messages

Those are handled by other protocols. ICMP mainly carries:

- Error information
- Control information
- Diagnostic information



ICMPv6 messages are divided into 4 main groups, depending on what job the message is doing.

Error messages: These are used when **something goes wrong with an IPv6 packet**.

Examples:

- Destination Unreachable
- Packet Too Big
- Time Exceeded
- Parameter Problem

Problem happens



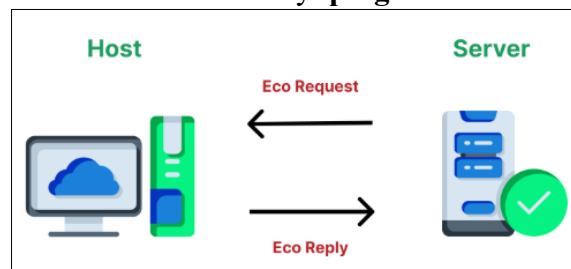
ICMPv6 sends an error message

Informational messages: These are used to **check or exchange information**, not necessarily because something went wrong.

The easiest examples are:

- Echo Request
- Echo Reply

Used by: **ping**



Neighbor Discovery messages: These are used by devices on the same local IPv6 network **to discover neighbors and routers**.

This group is related to the: ND Protocol

ND means: Neighbor Discovery

Examples of what it helps with:

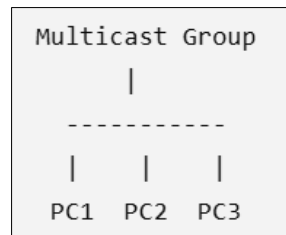
- Find a router
- Find another IPv6 device
- Learn link-layer information
- Check whether a neighbor is reachable

Easy idea: **ND Protocol** = *"Who is around me on this local IPv6 network?"*

Group Membership messages: These are used for IPv6 multicast groups. This group is related to MLD Protocol

- MLD means: Multicast Listener Discovery
- It helps routers know which devices want to receive traffic for a multicast group.

Example:



MLD helps answer: *"Which devices are members/listeners of this multicast group?"*

Q: Why ICMP is needed

Ans:

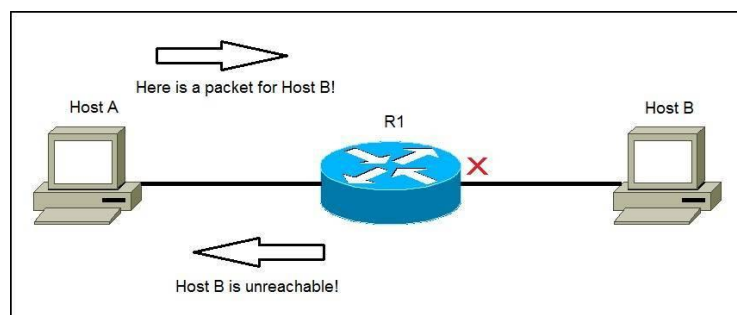
Suppose Computer A sends a packet to Computer B. But packet not sent.

Possible problems:

- Destination cannot be reached
- Packet is too large
- Packet travels for too many hops
- Something is wrong in the IP header

The sender needs to know about these problems. **ICMP is used to send that control/error information** back to the sender.

Here is an example:



Host A sends a packet to Host B. Because the Host B is down, the router will send an **ICMP Destination host unreachable** message to Host A, informing it that the destination host is unreachable, e.g.:

```
C:\>ping 192.168.8.11

Pinging 192.168.8.11 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.8.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

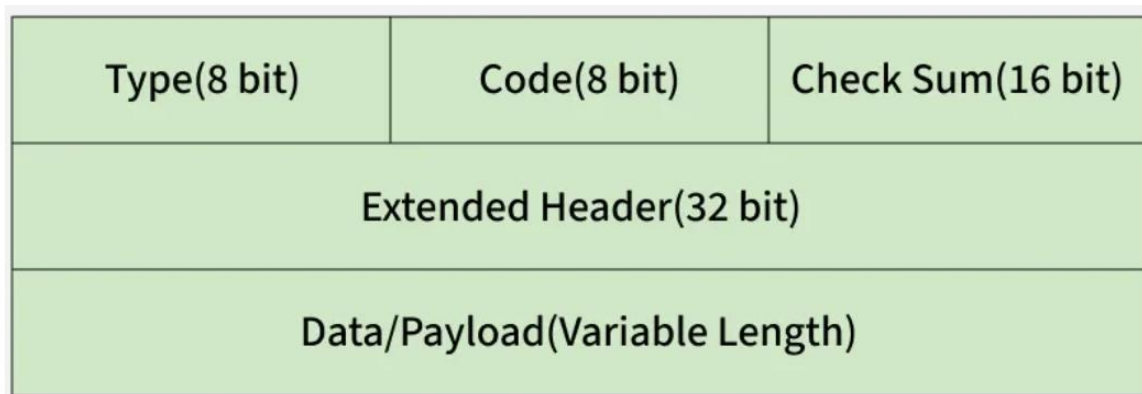
Q: What is the Internet Control Message Protocol Used for?

Ans:

1. A **network problem is diagnosed** and reported using the Internet Control Message Protocol (ICMP). When data does not arrive as expected, ICMP sends a message from the sender to the receiver. Ping and traceroute use ICMP messages as part of the diagnostic process to provide information on how data is being transmitted.
2. Some senders want to send a message to a destination, but the router is unable to do so. The router notifies the sender that I am unable to send the message to that destination.
3. ICMP is also commonly used for network diagnostics. Both traceroute and ping use ICMP. Traceroute and ping are messages sent when data is successfully transmitted. The physical routers that handled the data are included.
4. ICMP is also used to hurt network performance. ICMP floods, Smurf attacks, and ping of death attacks are used to overwhelm a device on the network and prevent it from functioning properly.

ICMP Packet Format

ICMP header comes after IPv4 and IPv6 packet header. In the ICMP packet format, the first 32 bits of the packet contain three fields:



ICMPv4 Packet Format

Type tells the main kind of message.

Code tells the more specific reason.

Checksum detects corruption.

More details: <https://www.geeksforgeeks.org/computer-networks/internet-control-message-protocol-icmp/>

ICMPv6 Error Messages

Q: What is an ICMPv6 error message?

Ans:

An ICMPv6 error message is a control message sent when a router or IPv6 host finds a problem while processing or delivering an IPv6 packet.

Q: How many main ICMPv6 error-message types are in this topic?

Ans:

The four basic ICMPv6 error-message types defined by RFC 4443 are:

Type	Error Message
1	Destination Unreachable
2	Packet Too Big
3	Time Exceeded
4	Parameter Problem

Q: What is the difference between Type and Code?

Ans:

Think of **Type** as the main category and **Code** as the exact reason inside that category.

Example: Type = 1 means: Destination Unreachable

But why is the destination unreachable? The Code tells that:

Code 0 → No route

Code 1 → Blocked by policy/firewall

Code 3 → Address unreachable

Code 4 → Port unreachable

So:

Type = What kind of problem?

Code = Why exactly did it happen?

The meaning of the Code always depends on the Type.

Q: What is Destination Unreachable?

Ans:

Destination Unreachable means the IPv6 packet could not be delivered to its intended destination.

- Its ICMPv6 Type is **Type = 1**
- A router or destination can generate this message when delivery fails for reasons other than congestion.
- Easy meaning: ***“I cannot deliver your packet to where you wanted it to go.”***

Q: Does every Destination Unreachable mean the same thing?

Ans:

No. That is exactly why the **Code field** exists. The original core codes are:

Code	Meaning
0	No route to destination
1	Communication administratively prohibited
2	Beyond scope of source address
3	Address unreachable
4	Port unreachable
5	Source address failed ingress/egress policy
6	Reject route to destination

Code 0 means the forwarding router checks its routing table but has no matching route and no usable default route for the destination.

Code 1 means the network may know how to reach the destination, but a security or administrative rule says, “Do not allow this traffic.”

Code 3 = I know where the network is, but I cannot reach that particular address.

Code 4 means the IPv6 destination host exists and is reachable, but the requested transport-layer port does not have a listening application. This commonly matters with UDP

Q: What does Code 2 mean: “Beyond scope of source address”?

Ans:

IPv6 addresses can have different scopes. For example, a Link-Local address: FE80::10 is valid only on the local link.

Suppose a device tries to use that Link-Local source address to communicate with a global destination through routers:

Source = FE80::10

Destination = 2001:db8:100::20

The destination is outside the valid scope of the source address.

That can produce:

Type = 1

Code = 2

RFC 4443 specifically gives a link-local source trying to reach a global-scope destination as an example.

Q: What is the difference between “Address Unreachable” and “Port Unreachable”?

Ans:

Address Unreachable → Cannot reach the destination interface/address itself.

Port Unreachable → Reached the destination host, but no application is listening on that port.

Example:

2001:db8::50 does not respond → Address Unreachable

2001:db8::50 responds, but UDP 9000 is closed → Port Unreachable

Type 2 error : Packet Too Big

Q: What is Packet Too Big?

Ans:

A Packet Too Big message is generated when a router cannot forward an IPv6 packet because the packet is larger than the MTU of the outgoing link.

Its values are:

Type = 2

Code = 0

Easy meaning: *“Your IPv6 packet is larger than this part of the path can carry.”*

Q: What is MTU?

Ans:

- MTU means: Maximum Transmission Unit
- It is the maximum packet size that a particular link can carry at the IPv6 layer without needing IPv6-layer fragmentation.

Example:

Ethernet link MTU = 1500 bytes

If an IPv6 packet is 1600 bytes

then $1600 > 1500$. so the packet cannot cross that link in its present form.

Q: What is Path MTU?

Ans:

A path may contain several links, and each link may have a different MTU. The Path MTU (PMTU) is the **smallest link MTU** anywhere along the path.

Example: PC ---- Link A ---- Router ---- Link B ---- Router ---- Link C ---- Server

Link A MTU = 1500

Link B MTU = 1400

Link C MTU = 1280

Therefore : $PMTU = \min(1500, 1400, 1280) = 1280$ bytes

So the largest packet that can cross the whole path without IPv6 fragmentation is: 1280 bytes

Q: What happens if an IPv6 router receives a packet that is larger than the outgoing link MTU?

Ans:

The router:

1. Does NOT fragment it.
2. Drops the packet.
3. Sends ICMPv6 Packet Too Big to the source.
4. Includes the outgoing-link MTU.

Q: Give a Packet Too Big mathematical example.

Ans:

Suppose:

IPv6 packet size = 1800 bytes

Outgoing MTU = 1500 bytes

Check: $1800 > 1500$.

Therefore the router cannot forward it.

Result:

ICMPv6 Type = 2

Code = 0

Reported MTU = 1500

The sender should adjust so subsequent packets fit the discovered path MTU.

Q: What is the minimum IPv6 link MTU?

Ans:

IPv6 minimum link MTU = **1280 bytes**

Q: What happens if a path has MTUs 9000, 1500, and 1280?

Ans:

PMTU = minimum of all link MTUs

Therefore:

$PMTU = \min(9000, 1500, 1280)$

= 1280 bytes

That is the largest IPv6 packet size that can traverse the entire path without source fragmentation.

Q: Why can blocking Packet Too Big messages cause problems?

Ans:

Path MTU Discovery relies on Packet Too Big messages. If these messages are blocked, a source may continue sending packets too large for some link, potentially creating a PMTU black-hole situation where a connection appears established but data transfer hangs.

Type 3 error : Time Exceeded

Q: What is the Time Exceeded message?

Ans:

Time Exceeded tells the sender that the packet could not complete its journey within the allowed forwarding/reassembly limit.

Its ICMPv6 Type is **Type = 3**

It has two core Codes:

Code 0 → Hop Limit exceeded in transit

Code 1 → Fragment reassembly time exceeded

Q: What is Time Exceeded Code 0?

Ans:

Code 0 occurs when the IPv6 **Hop Limit reaches zero** while the packet is still being forwarded.

- A router decreases the Hop Limit.
- If the router receives it as zero or decrements it to zero, the router discards the packet and sends Time Exceeded Code 0 back to the source.

Q: Suppose Hop Limit is 3, but the destination requires the packet to pass through 4 routers. What happens?

Ans:

- Start = 3
- Router 1 → 2
- Router 2 → 1
- Router 3 → 0
- At Router 3 Hop Limit becomes 0 so the packet is discarded before it can be forwarded further.

The router generates:

- ICMPv6 Time Exceeded
- Type = 3
- Code = 0

Q: Does “Time Exceeded” literally mean a stopwatch measured how long the packet travelled?

Ans:

For Code 0, no. The name can be misleading. The usual condition is Hop Limit reached 0. It is a hop-count limit, not ordinary elapsed clock time. However, Code 1 really is related to a reassembly timeout.

Q: What is Time Exceeded Code 1?

Ans:

- Code 1 means: Fragment reassembly time exceeded
- IPv6 fragments are reassembled at the destination. If enough fragments do not arrive to complete reassembly within the specified reassembly interval, the incomplete packet is abandoned
- When the relevant first fragment was received, an ICMP Time Exceeded Code 1 should be sent.
- RFC 8200 uses a 60-second reassembly timeout.

Example:

Type = 3
Code = 1

A source sends:
Fragment 1
Fragment 2
Fragment 3

Destination receives:
Fragment 1 ✓
Fragment 2 ✓
Fragment 3 ✗ missing

Type 4 error: Parameter Problem

Q: What is a Parameter Problem message?

Ans:

A Parameter Problem message means an IPv6 node found something invalid or unrecognized in the IPv6 header or extension headers and could not continue processing the packet.

- Its Type is: **Type = 4**
- The node discards the offending packet and can report the type and exact location of the problem.
- Easy meaning: ***“Something inside your IPv6 header is wrong.”***

Q: What are the three basic Parameter Problem codes?

Ans:

The core RFC 4443 codes are:

Code	Meaning
0	Erroneous header field encountered
1	Unrecognized Next Header type encountered

Q: What does Parameter Problem Code 0 mean?

Ans:

- Code 0 is the general case: Erroneous header field encountered
- The IPv6 node finds a field that is invalid and cannot properly process the packet.

Example: Suppose an IPv6 fragment has an invalid size/alignment condition that violates IPv6 fragmentation rules.

The receiving node may send:

- Parameter Problem
- Type = 4
- Code = 0

pointing to the field that caused the problem. IPv6 fragmentation rules define cases where Parameter Problem Code 0 is used.

Q: What does Parameter Problem Code 1 mean?

Ans:

Code 1 means: Unrecognized Next Header type

Remember that the IPv6 Next Header field tells what comes next:

6 → TCP
17 → UDP
58 → ICMPv6
44 → Fragment Header

Suppose a destination receives a Next Header value that it does not recognize or cannot process.

It may send:

Type = 4
Code = 1

Q: What does Parameter Problem Code 2 mean?

Ans:

Code 2 means: Unrecognized IPv6 option encountered

IPv6 extension headers can contain options. If an option is unknown and its processing rules require an error report, Parameter Problem Code 2 can be generated.

Q: What is the Pointer field in a Parameter Problem message?

Ans:

This is one of the most important parts of Parameter Problem.

The message contains a **Pointer** which gives the octet offset inside the original packet where the error was detected.

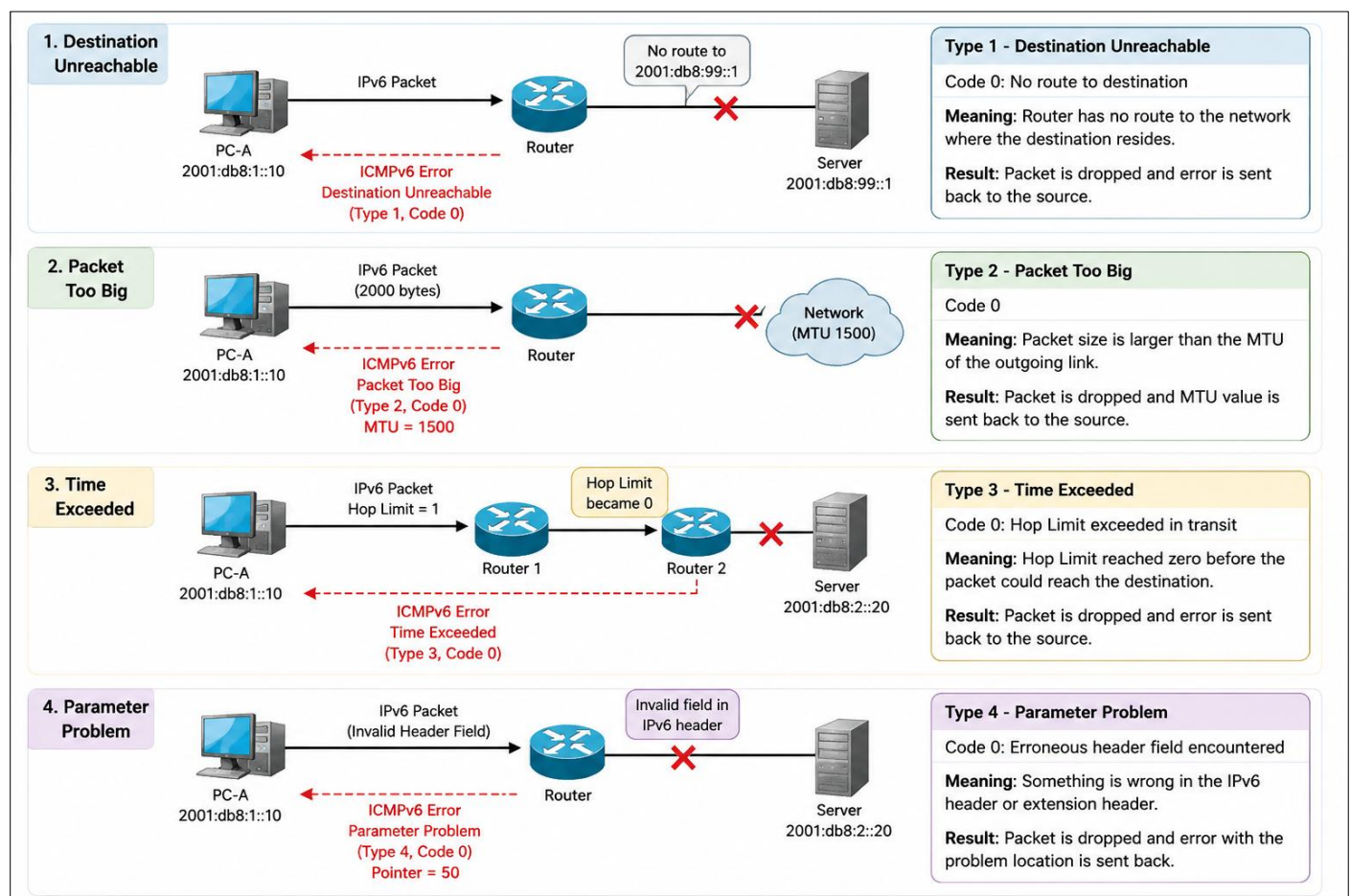
Think: Original IPv6 packet

0 1 2 3 ... 39 40 41 42 ...

^
|
error here

If: Pointer = 40 the problem was detected at byte offset 40 from the beginning of the original IPv6 packet.

Summary



ICMPv6 Informational Messages

Q: What is an ICMPv6 informational message?

Ans:

An ICMPv6 informational message is used to **exchange information or test communication**, rather than report that an IPv6 packet failed.

So compare:

ICMPv6 Error Message → "Something went wrong."

ICMPv6 Informational Message → "I want to check/exchange information."

The two basic informational messages defined here are:

Echo Request

Echo Reply

Q: How are ICMPv6 error and informational messages distinguished by Type values?

Ans:

ICMPv6 divides them using the Type field.

- Type 0–127 → Error messages
- Type 128–255 → Informational messages

That is why:

- Echo Request = Type 128
- Echo Reply = Type 129

Easy memory:

- Type < 128 → Error
- Type ≥ 128 → Informational

Q: What is an Echo Request message?

Ans:

An Echo Request is sent by one IPv6 device to another device to ask: ***“Can you receive my message and reply to me?”***

Its basic values are:

Type = 128
Code = 0

The destination can be a legal IPv6 destination address.

ICMPv6 nodes implement an Echo responder so that a received Echo Request can generate a corresponding Echo Reply.

Q: What is an Echo Reply message?

Ans:

An Echo Reply is the response sent after receiving an Echo Request.

Its values are:

Type = 129
Code = 0

The reply carries the same Identifier, Sequence Number, and Data from the corresponding request.

Q: How does the complete Echo Request–Echo Reply process work?

Ans:

Suppose:

PC-A IPv6 = 2001:db8:1::10

PC-B IPv6 = 2001:db8:2::20

PC-A wants to test whether PC-B is reachable.

First:

```
PC-A
|
| Echo Request
| Type = 128
v
PC-B
```

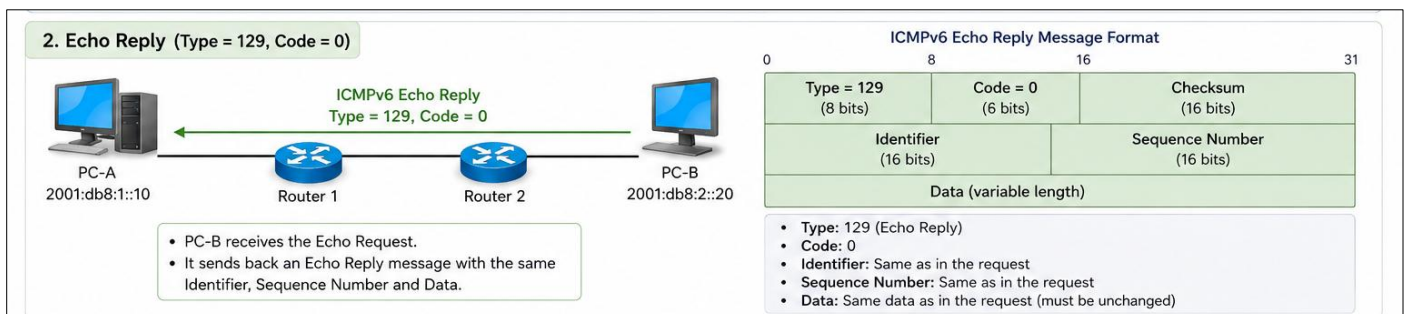
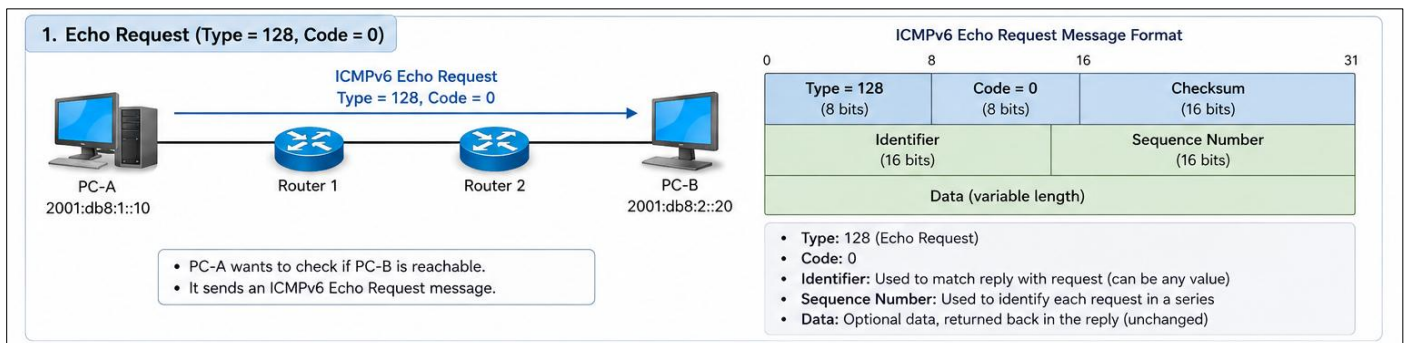
If PC-B receives and processes it, PC-B sends:

```
PC-B
|
| Echo Reply
| Type = 129
v
PC-A
```

Request = "Can you hear me?"

Reply = "Yes, I received you."

This request/reply mechanism is intended for diagnostic communication testing.



Q: What is the Checksum field?

Ans:

The Checksum is used to detect corruption in the ICMPv6 message.

Checksum → detects errors/corruption in the ICMPv6 message

Q: What is the Identifier field?

Ans:

The Identifier helps match an Echo Reply with the Echo Request that caused it.

Suppose two different Echo Request activities exist:

Request A

Identifier = 100

Request B

Identifier = 250

A reply arrives with:

Identifier = 100

Then the sender can associate it with: Request A

Easy meaning: Identifier → "Which Echo conversation/test does this belong to?"

Q: What is the Sequence Number?

Ans:

The **Sequence Number** helps distinguish one Echo Request from another within a series of requests. Like the Identifier, RFC 4443 defines it as a value used to help match replies with requests.

Suppose PC-A sends four requests:

Request 1 → Sequence = 1

Request 2 → Sequence = 2

Request 3 → Sequence = 3

Request 4 → Sequence = 4

Replies might be:

Reply → Sequence = 1

Reply → Sequence = 2

Reply → Sequence = 3

Reply → Sequence = 4

Now PC-A knows which reply corresponds to which request.

Q: What is the Data field?

Ans:

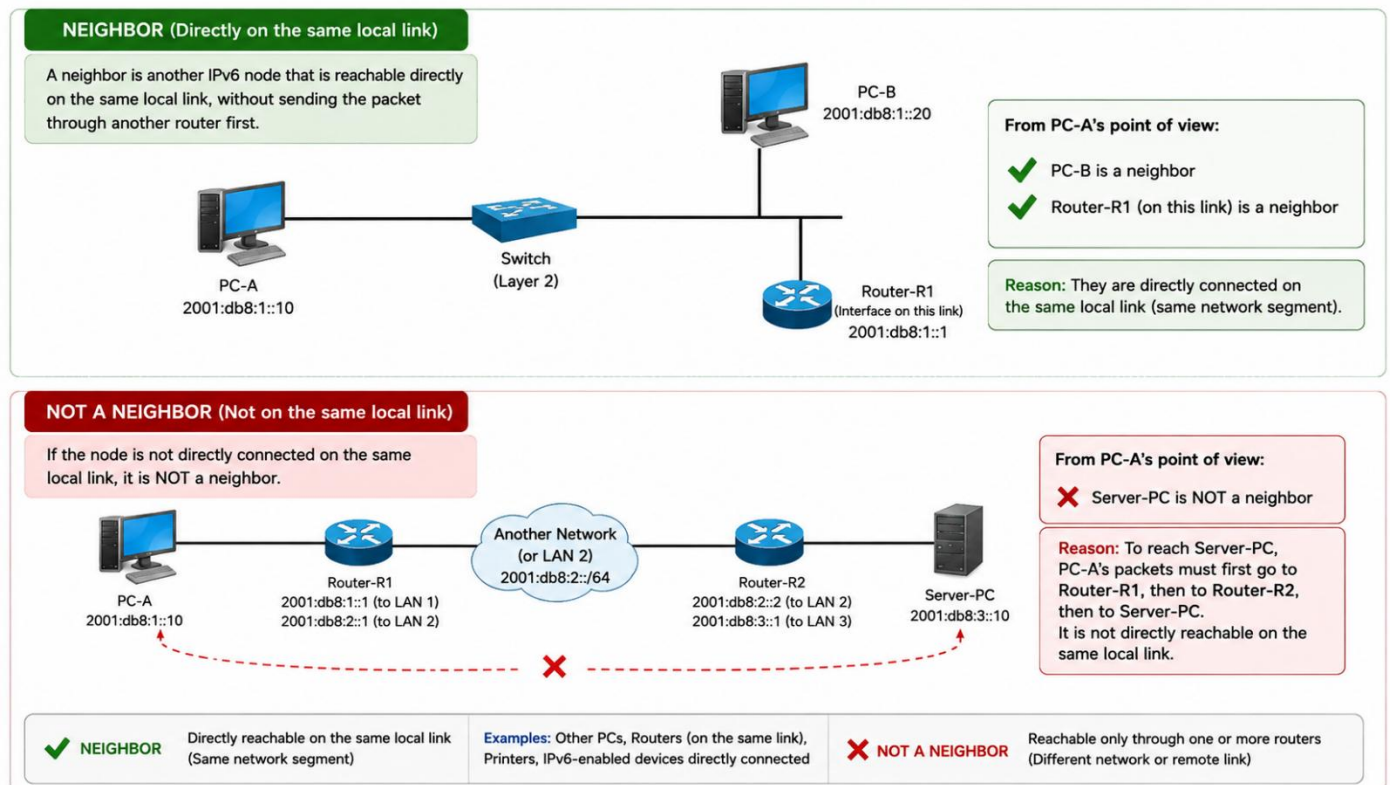
Echo Request can contain: zero or more octets of data. The Echo Reply must return the received Echo Request data unchanged.

For example:

Echo Request Data: HELLO123

Echo Reply Data: HELLO123 not: HELLO456

Neighbor Discovery Protocol



Q: What is Neighbor Discovery Protocol?

Ans:

Neighbor Discovery Protocol, usually written NDP or ND is an IPv6 mechanism that allows hosts and routers on the same link to discover and communicate with each other.

- ❖ The Neighbor Discovery Protocol is an IPv6 protocol, and it is essential for handling address autoconfiguration, resolution, and reachability detection.
- ❖ NDP operates at the link layer and uses Internet Control Message Protocol for IPv6 (ICMPv6) messages to handle various tasks such as address autoconfiguration, address resolution, and neighbor unreachability detection.

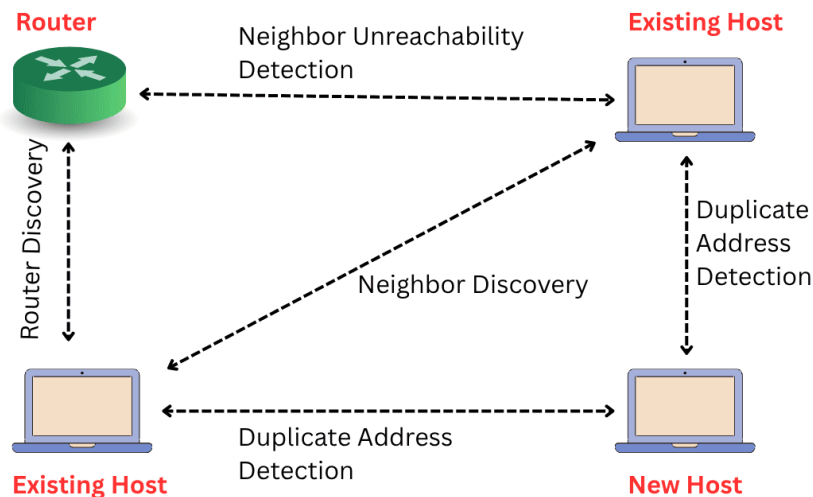
It performs several important jobs:

- Find routers
- Find neighboring devices
- Find a neighbor's link-layer address
- Check whether a neighbor is still reachable
- Learn network prefixes
- Help IPv6 address configuration
- Check duplicate addresses
- Learn a better first hop

The IPv6 Neighbor Discovery standard defines these functions, including address resolution, router discovery, Neighbor Unreachability Detection, Duplicate Address Detection, and Redirect processing.

Easy meaning: *ND helps an IPv6 device understand its local network.*

Neighbor Discovery Protocol



Q: Why does IPv6 need Neighbor Discovery?

Ans:

Imagine PC-A knows PC-B's IPv6 address:

- PC-A IPv6 = 2001:db8:1::10
- PC-B IPv6 = 2001:db8:1::20

But PC-A also needs the information required to actually deliver a frame to PC-B on the local link.

ND can perform address resolution:

IPv6 address
2001:db8:1::20
↓
Neighbor Discovery
↓
Link-layer address

ND also helps PC-A answer other questions:

- "Which router should I use?"
- "What IPv6 prefix is used here?"
- "Is 2001:db8:1::20 still reachable?"
- "Is the IPv6 address I want already being used?"

IPv6 ND combines functions that in IPv4 are spread across mechanisms such as ARP, Router Discovery, and Redirect.

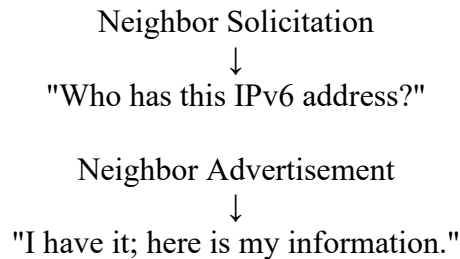
Q: Does IPv6 use ARP?

Ans:

Not for IPv6 address resolution. IPv4 commonly uses: ARP

IPv4 address → MAC/link-layer address

IPv6 uses **Neighbor Discovery**, particularly Neighbor Solicitation and Neighbor Advertisement:



Q: Describe the NDP message type.

Ans:

The Neighbor Discovery Protocol utilizes 5 ICMPv6 message types to communicate. Key messages include:

The first five are core ND;

The final two belong to the IND extension.

Message	Type	Purpose
Router Solicitation (RS)	133	Discover routers on the local network
Router Advertisement (RA)	134	Announce router presence and provide network information
Neighbor Solicitation (NS)	135	Ask about a neighbor
Neighbor Advertisement (NA)	136	Answer about a neighbor
Redirect	137	Inform hosts of a better first-hop router for a specific destination
Inverse Neighbor Solicitation	141	Known link-layer address → ask for IPv6 address
Inverse Neighbor Advertisement	142	Reply with IPv6 address information

Q: How Neighbor Discovery Protocol Works in IPv6?

Ans:

Here is a detailed step-by-step explanation of how the NDP protocol works.

Step 1. Address Autoconfiguration: When a device connects to an IPv6 network, it automatically configures its IPv6 address using Stateless Address Autoconfiguration (SLAAC).

Step 2. Router Discovery: The device sends a Router Solicitation (RS) message to discover routers. Routers respond with Router Advertisement (RA) messages, providing network prefix and configuration settings.

Step 3. Prefix Discovery: Routers provide the network prefix in their RA messages, allowing devices to configure their IPv6 addresses properly.

Step 4. Neighbor Discovery: To communicate with another device, a device sends a Neighbor Solicitation (NS) message to find the link-layer (MAC) address corresponding to an IPv6 address. The target device replies with a Neighbor Advertisement (NA) message, including its MAC address.

Step 5. Reachability Detection: Devices send periodic NS messages to check if neighbors' link-layer addresses are still valid. If no response is received, the device updates its neighbor cache accordingly.

Step 6. Duplicate Address Detection (DAD): Before assigning an IPv6 address, a device sends an NS message for the intended address. If no response is received, the address is safe to use. If a response is received, the address is considered duplicate.

Step 7. Neighbor Cache: NDP maintains a neighbor cache, storing mappings between IPv6 addresses and their corresponding MAC addresses. This cache improves efficiency by reducing the need for repeated NS messages.

Q: What is a Router Solicitation message?

Ans:

A Router Solicitation (RS) is sent by a host when it wants routers on the local network to send Router Advertisement information quickly.

Host → “Is there a router here? Please tell me about the network.”

Q: Why would a host send Router Solicitation?

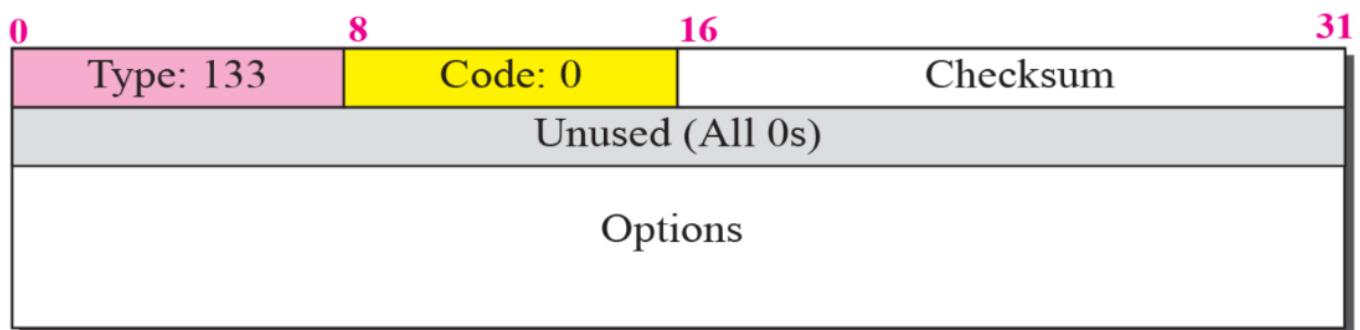
Ans:

Routers normally send Router Advertisements periodically. But imagine a laptop has just joined a Wi-Fi network. It does not want to wait unnecessarily for the next periodic advertisement.

So it can send: Router Solicitation asking routers to advertise immediately means "Router, please send me IPv6 network information."

Q: What fields are in Router Solicitation?

Ans:



Unused / Reserved

The Reserved field is unused and initialized to zero. Options may include the sender's link-layer address when appropriate.

Q: What is a Router Advertisement?

Ans:

A router sends a Router Advertisement to tell hosts that it is available and to provide IPv6 network configuration information.

Router → “Yes, I am a router. Here is the network information.”

Q: What is the relationship between RS and RA?

Ans:

RS = "Router, tell me about yourself/network."

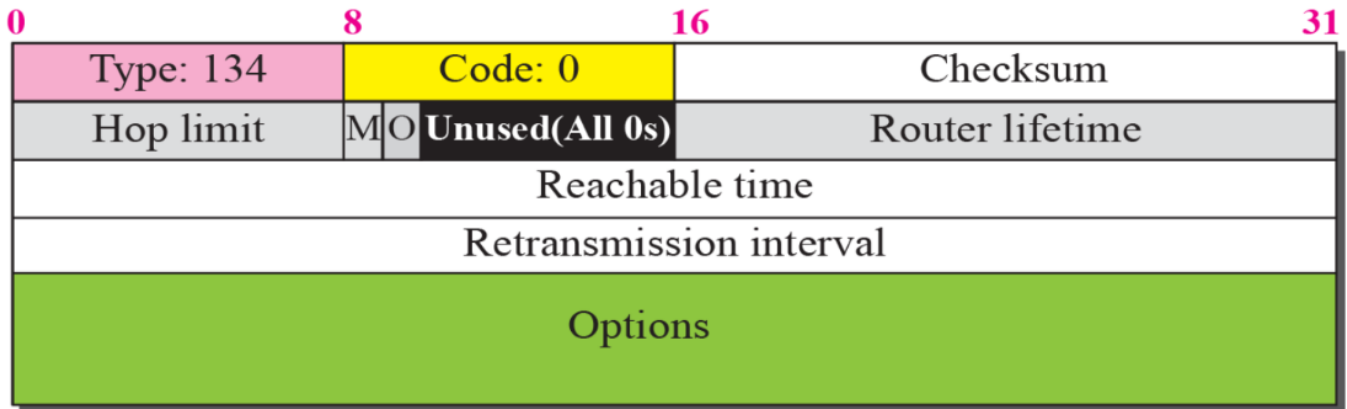
RA = "Here is the network/router information."

Q: What information can Router Advertisement provide?

Ans:

RA can provide important information such as:

- suggested Hop Limit,
- whether DHCPv6-related configuration is available,
- how long the router should be considered a default router,
- reachability/retransmission timing information,
- prefixes used on the link,
- and other options such as link parameters.



M means: Managed Address Configuration When set, it indicates that addresses are available through DHCPv6.

O means: Other Configuration. It indicates that other configuration information can be obtained through DHCPv6, such as certain network service information.

Router Lifetime tells a host how long that router should be considered usable as a default router. For example: Router Lifetime = 1800 seconds, means the host may consider that router a default-router candidate for that lifetime, subject to the protocol's normal update rules. A value of 0 says the router should not be used as a default router.

Q: What is a Neighbor Solicitation message?

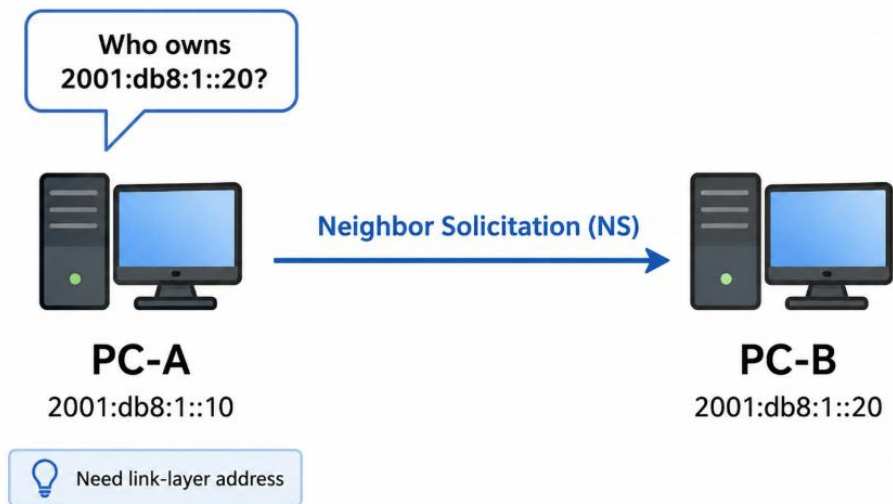
Ans:

A node sends a Neighbor Solicitation when it wants to find information about another IPv6 device on the same local link

Host A → "Who has this IPv6 address? Tell me your local link information."

It can be used to:

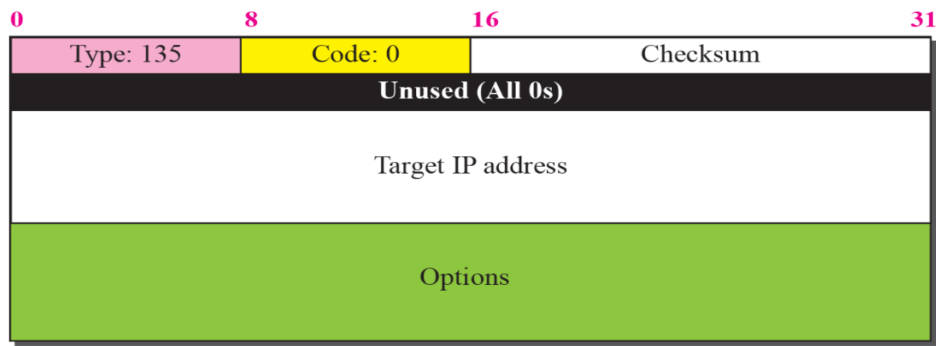
- discover a neighbor's link-layer address,
- check whether a known neighbor is still reachable,
- and perform Duplicate Address Detection.



Q: What fields are in Neighbor Solicitation?

Ans:

The important fields are:



The **Target Address** is the IPv6 address being queried

Q: What is the Target Address in Neighbor Solicitation?

Ans:

It is the IPv6 address of the neighbor being asked about.

Example: PC-B IPv6 = 2001:db8:1::20

Then NS can contain target Address = 2001:db8:1::20

The Target Address cannot be a multicast address.

Q: What is Neighbor Advertisement?

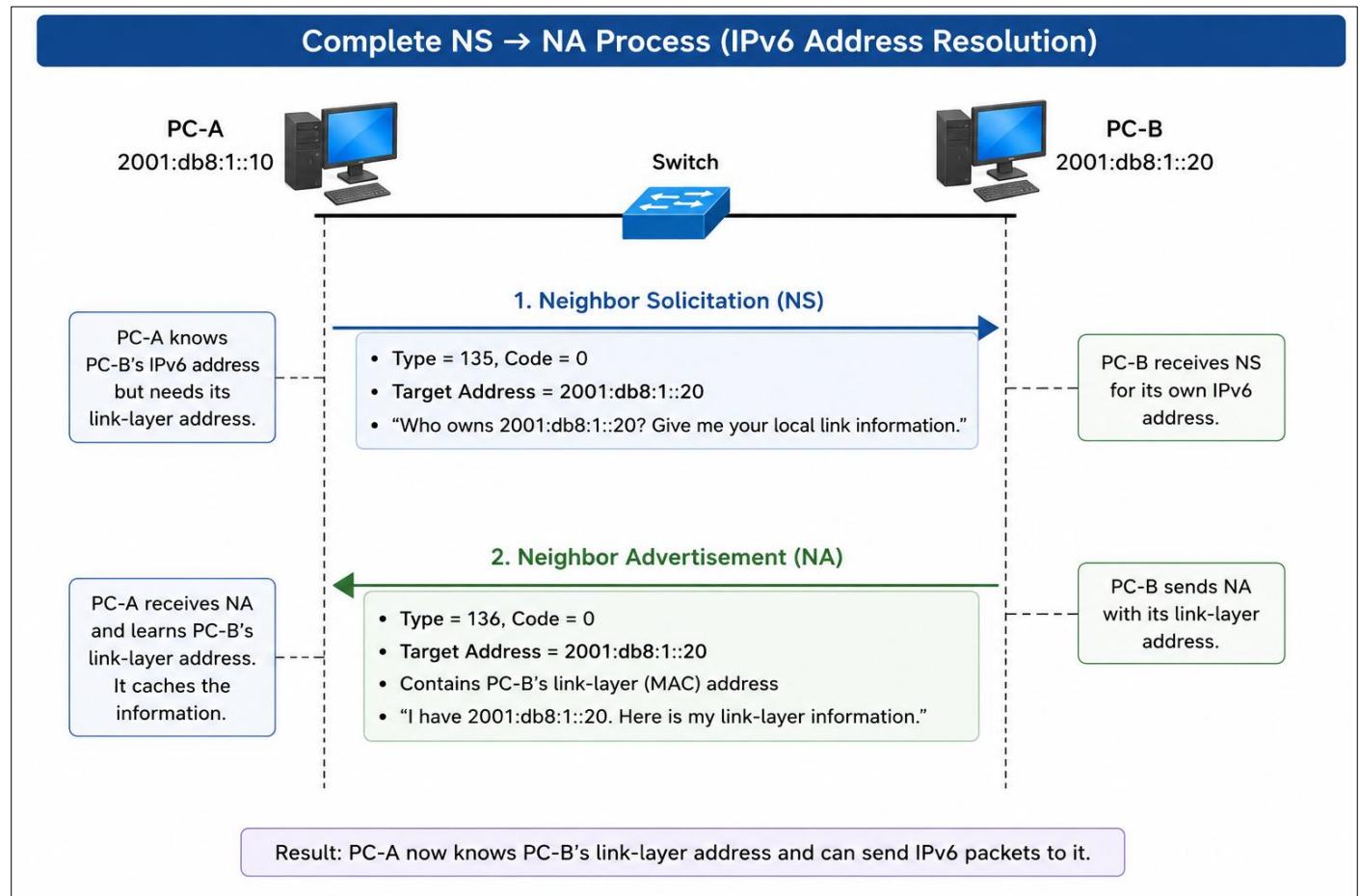
Ans:

A Neighbor Advertisement (NA) normally answers a Neighbor Solicitation.

NS = "Who has this IPv6 address?"

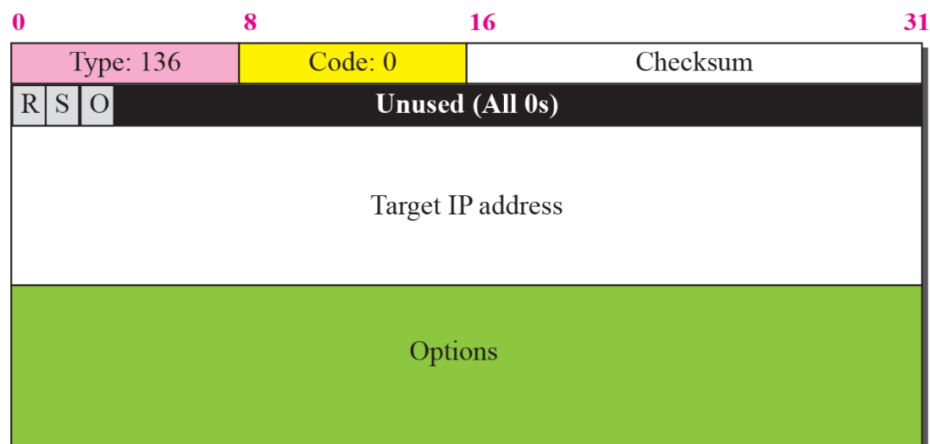
NA = "I have it."

A node can also send an unsolicited Neighbor Advertisement to announce changed link-layer information.



Q: What fields are in Neighbor Advertisement?

Ans:



R is the **Router flag**.

R = 1 → sender is a router

It helps nodes maintain correct information about whether a neighbor acts as a router.

S is the **Solicited flag**.

If S = 1 the Neighbor Advertisement was sent in response to a Neighbor Solicitation from the destination.

S = Solicited response

O is the **Override flag**.

When appropriate, it tells the receiver whether the advertisement's link-layer information should override/update an existing Neighbor Cache entry.

Q: What is Duplicate Address Detection?

Ans:

Duplicate Address Detection (DAD) checks whether an IPv6 address a device wants to use is already being used by another device.

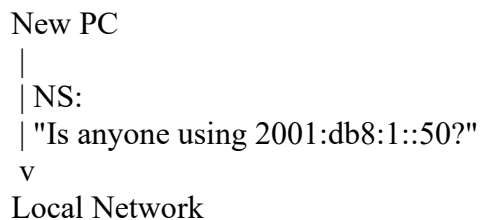
Before using a tentative address, the device essentially asks: *"Is anybody already using this IPv6 address?"* Neighbor Solicitation is used for this purpose.

Example.

Suppose a new PC wants: 2001:db8:1::50

Before fully using it, it sends an NS targeting that tentative address.

Conceptually:



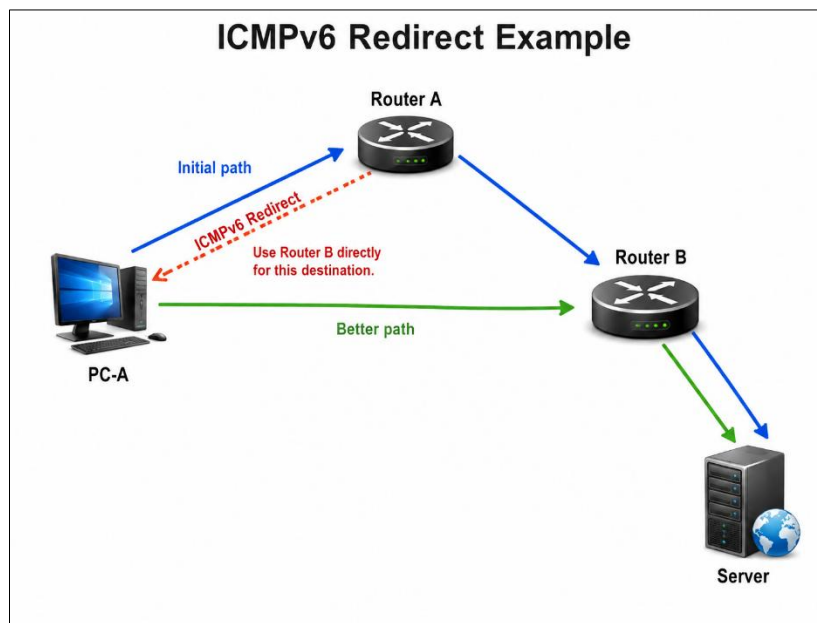
- If nobody indicates ownership: Address appears available
- If another node already uses it and responds appropriately: Duplicate detected

For DAD, the NS uses the unspecified source address :: because the tentative address has not yet been confirmed for use. (

Q: What is an IPv6 Redirect message?

Ans:

A **Redirect** is sent by a router to tell a host: *“There is a better first hop for this destination.”*



0	8	16	31
Type: 137	Code: 0	Checksum	
Reserved			
Target (router) IP address			
Destination IP address			
Options			

Q: What important addresses are inside Redirect?

Ans:

Target (Router) IP Address

Destination IP Address

In standard ND:

- **Target Address** identifies the better next hop.
- **Destination Address** identifies the destination for which the host should use that target.

Q: Can a normal host send Redirect messages?

Ans:

No. Redirect messages are sent by routers; hosts do not send IPv6 ND Redirect messages.

Q: What does “Inverse Neighbor Discovery” mean?

Ans:

Normal Neighbor Discovery commonly solves:

Known IPv6 address



Find link-layer address

Inverse Neighbor Discovery (IND) solves the reverse type of problem:

Known link-layer address



Find IPv6 address(es)

Easy memory:

Normal ND:

IPv6 → Link-layer

Inverse ND:

Link-layer → IPv6

Q: What is Inverse Neighbor Solicitation?

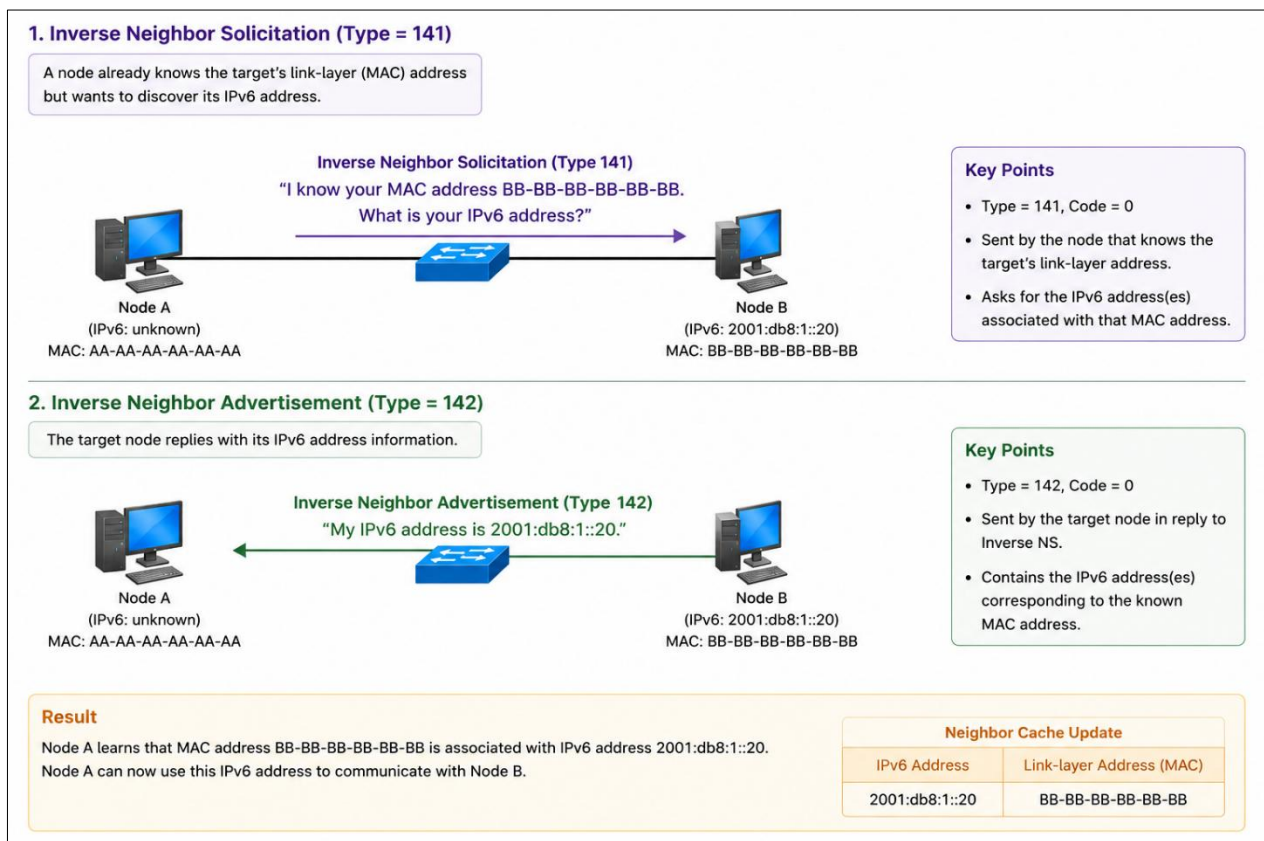
Ans:

Inverse Neighbor Solicitation asks: ***“I know your link-layer identity; what IPv6 address or addresses belong to you?”***

Q: What is Inverse Neighbor Advertisement?

Ans:

It is the response to an Inverse Neighbor Solicitation.



Group Membership Messages

Q: What does “group membership” mean?

Ans:

It simply means: Which devices are interested in receiving traffic for a particular multicast address?

Suppose the multicast group is: FF3E::100

and:

PC-A → wants FF3E::100

PC-B → wants FF3E::100

PC-C → does NOT want FF3E::100

Then:

Members/listeners:

PC-A

PC-B

PC-C is not interested.

The router needs to know this so **that it does not unnecessarily forward multicast traffic onto links** where nobody wants it. MLD exists specifically so IPv6 routers can learn which multicast addresses have interested listeners on their directly connected links.

Q: What is a multicast listener?

Ans:

A multicast listener is an IPv6 node that wants to receive packets sent to a particular multicast address.

For example:

Video multicast address: FF3E::500

If PC-A wants to receive that stream:

PC-A

↓

Listening to FF3E::500

then PC-A is a multicast listener for that group.

A listener can be a host, and a router can also itself be a listener.

MLD basics

Q: What is MLD?

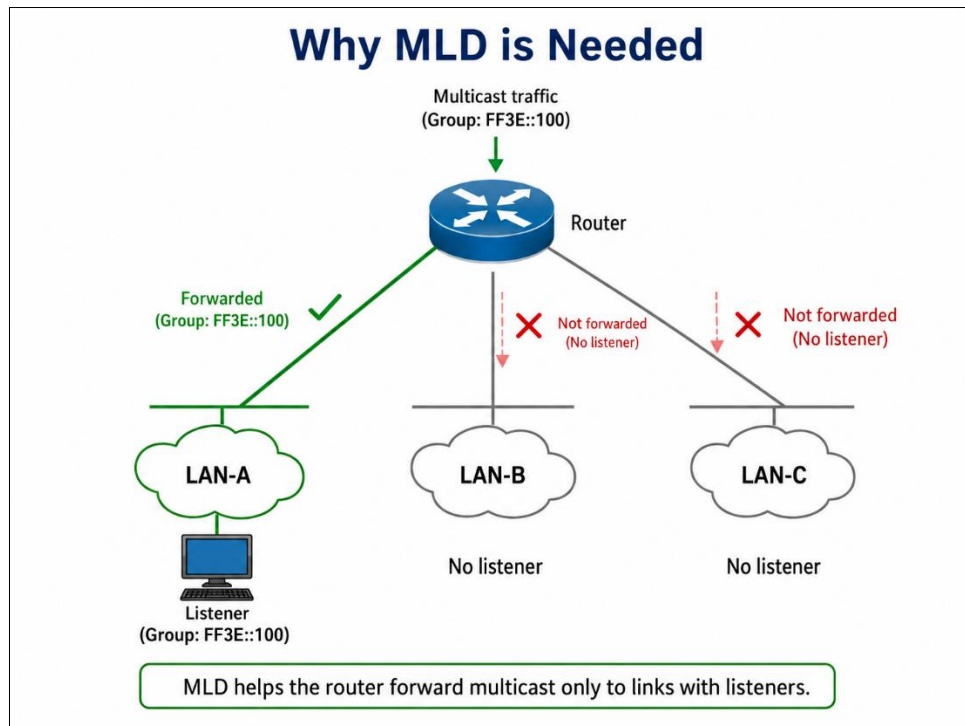
Ans:

MLD means: Multicast Listener Discovery. It is the IPv6 protocol used by routers to discover: *"Are there multicast listeners on this link?"* and *"Which multicast groups are they interested in?"*

The router uses this information so multicast traffic is forwarded only toward links that actually have interested listeners.

The important versions are:

1. MLDv1
2. MLDv2



Q: What are the two main MLDv2 messages?

Ans:

Message	Type	Direction
Multicast Listener Query	130	Router → listeners
MLDv2 Multicast Listener Report	143	Listener → routers

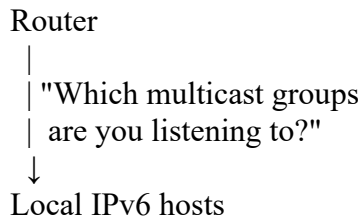
Membership Query

Q: What is a Multicast Listener Query?

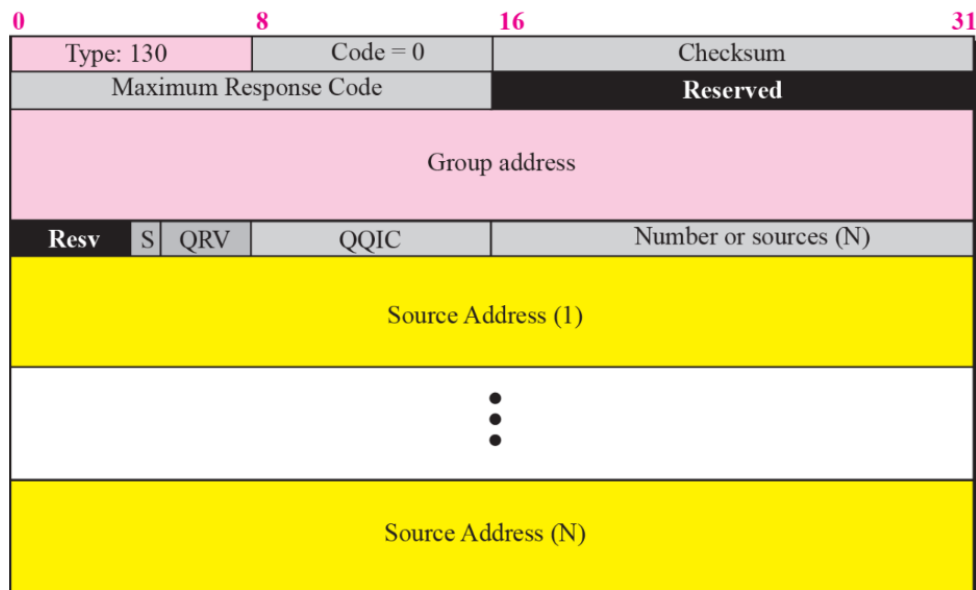
Ans:

A Multicast Listener Query is sent by the multicast router to ask listeners about their multicast interests.

Conceptually:



The listeners then respond with reports describing their multicast listening state.



Type = 130 identifies the packet as a Multicast Listener Query

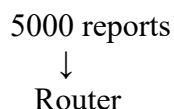
For MLDv2 Query Code = 0. The standard says the sender initializes it to zero and receivers ignore it.

Q: What is the Maximum Response Code?

Ans:

It tells listeners the **maximum amount of time they may wait** before sending a Report in response to the Query.

Imagine 5,000 listeners all replying at exactly the same moment:



That could create a burst. Instead, listeners spread their responses within the allowed response period.

Q: What does Number of Sources mean?

Ans:

It tells how many source IPv6 addresses follow in the Query.

- For General Query and Multicast Address Specific Query normally: $N = 0$
- For Multicast Address + Source Specific Query it can be: $N > 0$

Q: What is QQIC?

Ans:

QQIC means: Querier's Query Interval Code. It tells other routers/listeners about the querier's Query Interval.

Simple idea:

QQIC

↓

"Approximately how often is the querier sending its regular Queries?"

If:

QQIC < 128

then:

QQI = QQIC seconds

Q: Give a simple QQIC calculation.

Ans:

Suppose: QQIC = 60

Since: $60 < 128$

we use: QQI = 60 seconds. So the query interval represented is: 60 seconds

Q: What happens if QQIC ≥ 128 ?

Ans:

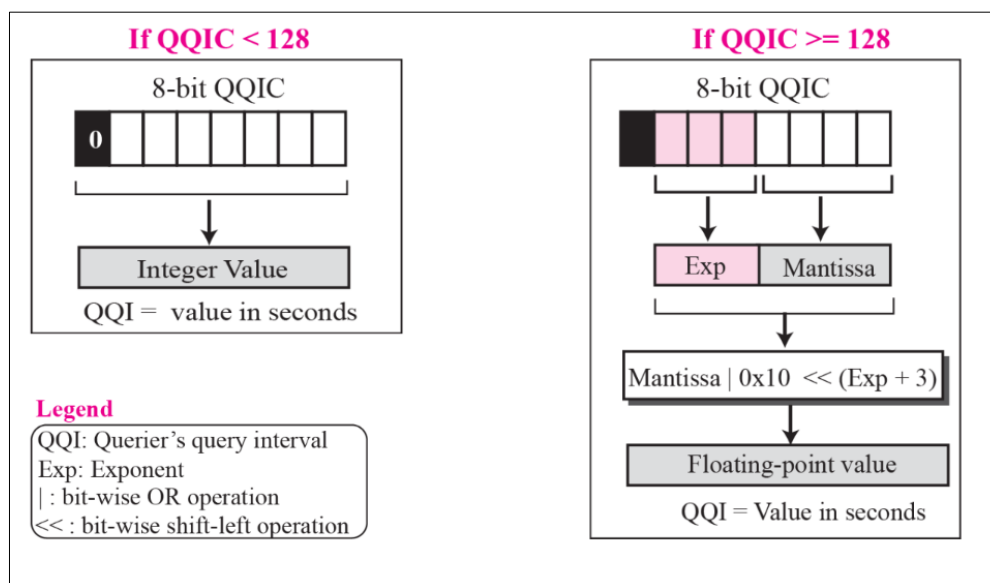
Then the value uses an exponent/mantissa form:

|1| exp | mant|

and:

QQI = (mant OR 0x10) << (exp + 3)

seconds.



Q: Give a QQIC floating-point example.

Ans:

Suppose the encoded bits after the leading 1 give:

exp = 001

mant = 0010

Then: mant OR 0x10 = 0010 OR 10000 = 10010₂ = 18

and:

$$\text{exp} + 3 = 1 + 3 = 4$$

Therefore:

$$\text{QQI} = 18 \ll 4 = 18 \times 16 = 288 \text{ seconds}$$

So:

Query Interval = 288 seconds

Membership Report

Q: What is an MLDv2 Multicast Listener Report?

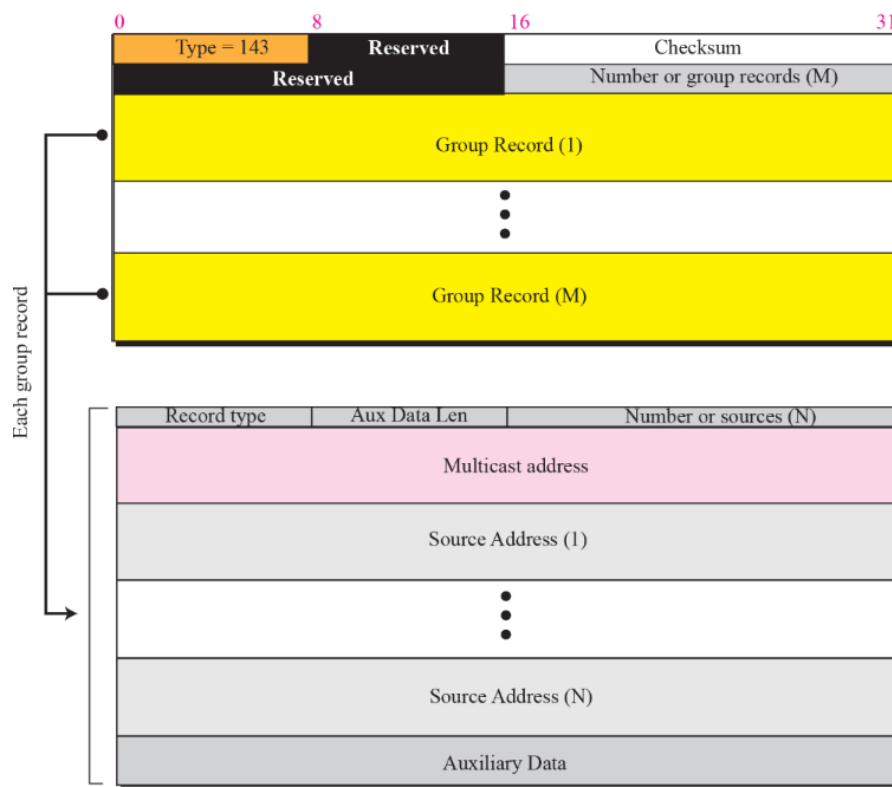
Ans:

A Report is sent by an IPv6 listener to tell neighboring routers about its current multicast listening state or a change in its multicast listening state

Easy idea:

Router: "Who is listening?"

Host: "I am listening to these groups/sources."



Q: What is a Multicast Address Record?

Ans:

One Report can contain information about multiple multicast groups. Each Multicast Address Record describes the listener's state for one multicast address.

Example:

Report

Record 1

→ FF3E::100

Record 2

→ FF3E::200

Record 3

→ FF3E::300

Therefore a single MLDv2 Report can carry information about several multicast groups.

Q: What is source filtering?

Ans:

Suppose a multicast group is: FF3E::100 and there are three possible senders:

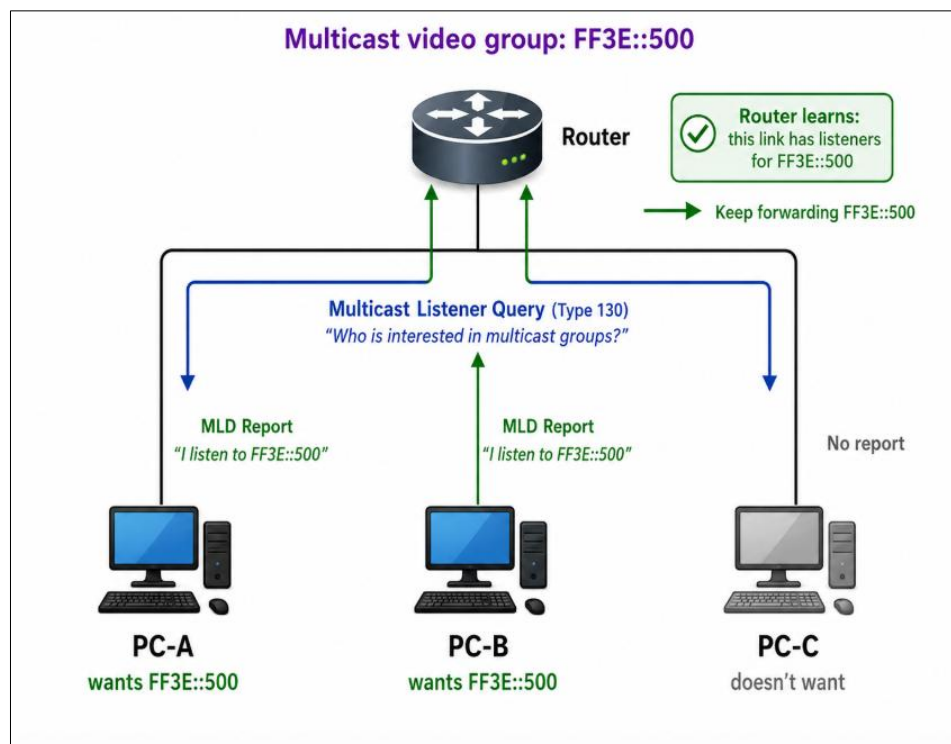
Source A = 2001:db8:A::1

Source B = 2001:db8:B::1

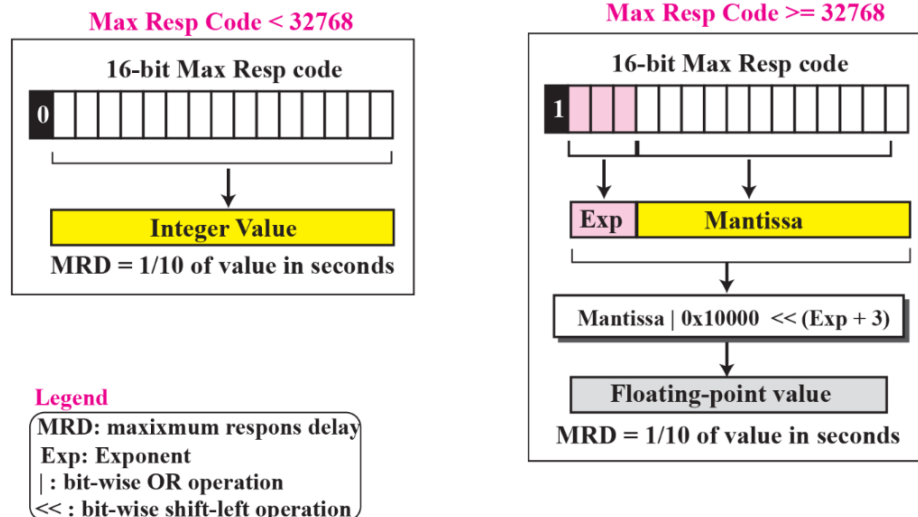
Source C = 2001:db8:C::1

A listener might say: "I want FF3E::100, but only from Source A." That is **source filtering**.

MLDv2 allows the listener to specify which senders it wants or does not want for a multicast group.



Calculation questions



Q: A General Query has Maximum Response Code = 5000 under the current MLDv2 standard. What is the Maximum Response Delay?

Ans:

Because: $5000 < 32768$

use MRD = Code milliseconds

Therefore: MRD = 5000 ms

Convert: 5000 ms = 5 seconds

Answer: Maximum Response Delay = 5 seconds

Q: QQIC = 100. What is the Query Interval?

Ans:

Since:

$100 < 128$

therefore:

QQI = 100 seconds

Q: A Query contains 4 source addresses. What does Number of Sources contain?

Ans:

N = 4

because Number of Sources tells how many IPv6 source-address fields follow.

Q: What is the difference between Neighbor Discovery and MLD?

Ans:

Feature	Neighbor Discovery	MLD
Main job	Discover/manage local neighbors and routers	Manage multicast listeners
Example messages	RS, RA, NS, NA	Query, Report
ICMPv6 based	Yes	Yes
Main question	"Who/what is on my local link?"	"Who wants this multicast traffic?"
Hop Limit behavior	Core ND expects 255	MLD uses 1

Scenario Based Questions

Q: A router receives a packet for 2001:db8:99::5, but there is no matching route and no default route. Which ICMPv6 error is appropriate?

Ans:

Destination Unreachable

Type = 1

Code = 0

Reason: No route to destination

Q: A firewall deliberately blocks communication to an IPv6 server. Which error/code fits?

Ans:

Destination Unreachable

Type = 1

Code = 1

because communication is administratively prohibited.

Q: A computer uses FE80::10 as its source address and tries to communicate with a distant global IPv6 address through routers. What may happen?

Ans:

A Link-Local source cannot be used beyond its local-link scope.

The appropriate error can be:

Destination Unreachable

Type = 1

Code = 2

meaning: Beyond scope of source address

Q: An IPv6 server is reachable, but a UDP datagram is sent to port 7000 and no service is listening there. Which error?

Ans:

Destination Unreachable

Type = 1

Code = 4

Reason: Port unreachable

Q: A 2000-byte IPv6 packet reaches a router whose next outgoing link has MTU 1500. What happens?

Ans:

$2000 > 1500$

The router drops it and sends:

Packet Too Big

Type = 2

Code = 0

MTU = 1500

The router does not fragment the IPv6 packet.

Q: A packet keeps cycling because Router A sends it to Router B and Router B sends it back to Router A. What eventually stops it?

Ans:

Its Hop Limit continually decreases:

$5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$

When it reaches zero, the router drops it and generates:

Time Exceeded

Type = 3

Code = 0

This is exactly one reason Time Exceeded exists.

Q: An IPv6 destination receives fragments 1, 2, and 4, but fragment 3 never arrives. Reassembly cannot finish before the timer expires. Which error?

Ans:

Time Exceeded

Type = 3

Code = 1

because Fragment reassembly time exceeded

Q: A destination encounters an unknown Next Header value while parsing an IPv6 packet. Which error?

Ans:

Parameter Problem

Type = 4

Code = 1

The Pointer field indicates where the problematic Next Header value was found.

Q: A node finds an invalid field inside an IPv6 header, but the issue is not specifically an unknown Next Header or unknown option. Which error?

Ans:

Parameter Problem

Type = 4

Code = 0

because: Erroneous header field encountered

Q: PC-A sends an ICMPv6 packet with Type 128, Code 0. What is it?

Ans:

Type 128 → Echo Request. Therefore PC-A is sending an informational/diagnostic request. ([RFC Editor](#))

Q: PC-B sends back Type 129, Code 0. What is it?

Ans:

Type 129 → Echo Reply. It is responding to an Echo Request

Q: A new laptop joins an IPv6 Wi-Fi network and wants a router to immediately provide network information. What message does it send?

Ans:

Router Solicitation

Type = 133

Code = 0

The router can answer with:

Router Advertisement

Type = 134

Q: A router periodically tells all local IPv6 hosts that it exists and advertises prefixes and configuration information. Which message is this?

Ans:

Router Advertisement

Type = 134

Code = 0

Q: PC-A knows 2001:db8:1::20 but does not know the neighbor's link-layer address. What message should PC-A use?

Ans:

Neighbor Solicitation

Type = 135

Code = 0

Then the target can respond with:

Neighbor Advertisement

Type = 136

Code = 0

Q: PC-B receives an NS for its own IPv6 address. Which message normally answers it?

Ans:

Neighbor Advertisement

Type = 136

Easy:

NS → question

NA → answer

Q: A computer wants to use 2001:db8:1::50 but first needs to verify that nobody else already uses it. Which mechanism is involved?

Ans:

Duplicate Address Detection

It uses Neighbor Solicitation targeting the tentative address.

Q: A host has been communicating with a router but now needs to verify that the router is still reachable. Which ND function is involved?

Ans:

Neighbor Unreachability Detection

NS/NA communication can help verify reachability.

Q: Router A realizes that PC-A should use Router B as a better first hop to a destination. Which message can Router A send?

Ans:

Redirect

Type = 137

Code = 0

Q: A node already knows another node's link-layer address but wants to learn its IPv6 address. Which mechanism is this?

Ans:

Inverse Neighbor Discovery

It begins with:

Inverse Neighbor Solicitation

Type = 141

and can receive:

Inverse Neighbor Advertisement

Type = 142

Q: You inspect an IPv6 packet and see:

Next Header = 58

ICMPv6 Type = 130

What is it?

Ans:

First:

Next Header 58 → ICMPv6

Then:

Type 130 → Multicast Listener Query

Therefore it is an: MLD Query

Q: You see ICMPv6 Type 143. What does it mean?

Ans:

Type 143 → MLDv2 Multicast Listener Report

It is being used to report multicast listening state or a change in that state to neighboring multicast routers.